extrapolation is carried out between the nearest mode points:
.1 Zone A uses mode points 5, 1, 6 and 2. Zone A may extend above the torque line from mode point 5 and mode point 1 or beyond the speed line from mode point 1 to mode point 5.
.2 Zone B uses mode points 6, 2, 7 and 3. Zone B may extend beyond the speed line from mode point 2 to mode point 3.
.3 The applicant may request that the Administration exclude operating points from the limit area of the not-to-exceed zone screening if the applicant can demonstrate that the engine is not capable of operating at steady state at those points when installed on a ship. Otherwise, the not-to-exceed zone consists of the entire limit area of the not-to-exceed zone.
5.2 Determine if the checkpoint is in Zone A or Zone B by determining if the

---CHUNK---

SOURCE: MEPC.397(83).pdf
on a ship. Otherwise, the not-to-exceed zone consists of the entire limit area of the not-to-exceed zone.
5.2 Determine if the checkpoint is in Zone A or Zone B by determining if the checkpoint torque, , is higher or lower than the torque on the boundary between Zone A and Zone B (75% load line) for the checkpoint speed.
 = 6 + ( ) (62)() (12)
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 19 -
where:
 = Torque at checkpoint speed on a straight line between mode point 6 and mode point 2 (75% load line)
6 = Torque at mode point 6 (75% of torque at intermediate speed)
2 = Torque at mode point 2 (75% of torque at rated speed)
 = Checkpoint speed
 = Intermediate speed
 = Rated speed
5.3 Determine the interpolated/extrapolated NOx value at desired checkpoint:
.1 If the checkpoint torque, , is greater than , use equation (13) and (14) for the interpolation or extrapolation.
 = + ( ) ()() (13)
with:
 = 5 + ( ) (51)() (14)
where:

---CHUNK---

SOURCE: MEPC.397(83).pdf
.1 If the checkpoint torque, , is greater than , use equation (13) and (14) for the interpolation or extrapolation.
 = + ( ) ()() (13)
with:
 = 5 + ( ) (51)() (14)
where:
1 = Torque at mode point 1 (100% of torque at rated speed)
5 = Torque at mode point 5 (100% of torque at intermediate speed)

= Torque at checkpoint speed on a straight line between mode
point 6 and mode point 2 (75% load line) from formula (12)
= Torque at checkpoint speed on a straight line between mode
point 5 and mode point 1 (100% load line) from formula (14)
= Torque at checkpoint
= Interpolated NOx at checkpoint speed on 100% load line
= Interpolated NOx at checkpoint speed on 75% load line
= Checkpoint speed
= Intermediate speed
= Rated speed
.2 If the checkpoint torque, , is less than use equation (15) and
(16) for the interpolation or extrapolation.
= + ( ) ()() (15)
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 20
with:

---CHUNK---

SOURCE: MEPC.397(83).pdf
= + ( ) ()() (15)
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 20
with:
= 7 + ( ) (73)() (16)
where:
3 =Torque at mode point 3 (50% of torque at rated speed)
7 =Torque at mode point 7 (50% of torque at intermediate speed)
=Torque at checkpoint speed on a straight line between mode
point 6 and mode point 2 (75% load line) from formula (12)
=Torque at checkpoint speed on a straight line between mode
point 7 and mode point 3 (50% load line) from formula (16)
=Torque at checkpoint
=Interpolated NOx at checkpoint speed on 75% load line
=Interpolated NOx at checkpoint speed on 50% load line
= Checkpoint speed
=Intermediate speed
=Rated speed
5.4 Determine the not-to-exceed emission limit value at checkpoint:
.1 For Tier II
The not-to-exceed emission limit value is given by formula (17):
= 1.2 (17)
where:
= Not-to-exceed emission limit value at checkpoint
= Interpolated NOx value at power
.2 For Tier III

---CHUNK---

SOURCE: MEPC.397(83).pdf
.1 For Tier II
The not-to-exceed emission limit value is given by formula (17):

 = 1.2 (17)
where:
 = Not-to-exceed emission limit value at checkpoint
 = Interpolated NOx value at power
.2 For Tier III
The not-to-exceed emission limit value shall be either set by 3.1.4
of this Code or as determined in accordance with 5.3, whichever is
the lower:
 is the lower of or
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 21 -
with:
 = + 0.25 (18)
 = 1.5 (19)
where:
 = Interpolated NOx value at checkpoint
 = NOx cycle limit for engine
  = the maximum mode point value according to 3.1.4 of
this Code
6 Method to address discontinuity in the operation zone due to an
auxiliary control device
6.1 For each approved auxiliary control device, where there is operation in the
not-to-exceed zone that causes a discontinuity in the NO x emissions it can be
necessary to introduce additional proxy mode points to account for the discontinuity

---CHUNK---

SOURCE: MEPC.397(83).pdf
6.1 For each approved auxiliary control device, where there is operation in the
not-to-exceed zone that causes a discontinuity in the NO x emissions it can be
necessary to introduce additional proxy mode points to account for the discontinuity
in the area of engine operation where that auxiliary control device is active.
6.2 There will be two or more proxy mode points to cover the action of an
auxiliary control device.
6.3 is calculated in the same manner as 4.2 using the proxy points where
necessary in the interpolation.
6.4 Use good engineering judgement that may include simulation or in -house
testing to determine the appropriate NOx level and location of the proxy points.
6.5 The engine power of the proxy mode points may overlap to account for
hysteresis that may occur as a result of approaching the points from rising or falling
power. The overlap should also take into account any variation in the operating point
of the auxiliary control device based on engine speed.
6.6 Include the proxy mode points as part of the auxiliary control device
documentation supplied to the Administration in the NOx certification pack."

---CHUNK---

SOURCE: MEPC.397(83).pdf
of the auxiliary control device based on engine speed.
6.6 Include the proxy mode points as part of the auxiliary control device
documentation supplied to the Administration in the NOx certification pack."
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION

OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)

---CHUNK---

SOURCE: MEPC.398(83).pdf
RESOLUTION MEPC.398(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)RESOLUTION MEPC.398(83)
(adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES
(NOx TECHNICAL CODE 2008)
(Certification of an engine subject to substantial modification or being certified to a
Tier to which the engine was not certified at the time of its installation)
THE MARINE ENVIRONMENT PROTECTION COMMITTEE
RECALLING Article 38(a) of the Convention on the International Maritime Organization
concerning the functions of the Marine Environment Protection Committee conferred upon it
by international conventions for the prevention and control of marine pollution from ships,
RECALLING ALSO article 16 of the International Convention for the Prevention of Pollution
from Ships, 1973, as modified by the Protocols of 1978 and 1997 relating thereto (MARPOL),
which specifies the amendment procedure and confers upon the appropriate body of